

IA: An Investigation to Investigate the Effect of Concentration on Rate of Reaction

Research Question

*To what extent does concentration of urease affect the rate of reaction between urease from
Canavalia ensiformis and urea?*

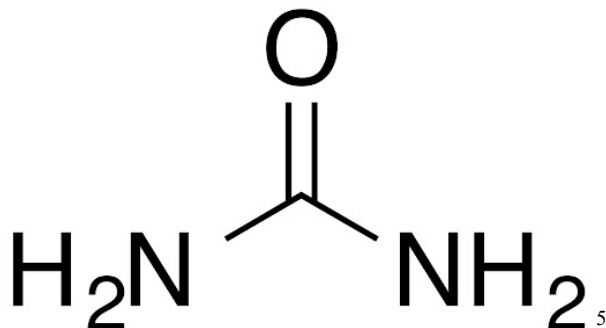
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Background

Nitrogen is one of the most abundant gases in the atmosphere and is considered the building blocks for all life and particularly plants¹. However, despite its plentifulness, plants are unable to utilize nitrogen in its gaseous state. Thus, legumes and other bacteria like *Rhizobium* help fix gaseous nitrogen into useable ammonia which can be taken up by plants, playing a crucial role in the nitrogen cycle². Ammonia won't remain stable in plants for long periods of time due to its toxicity. Accumulation of ammonia disrupts cellular processes like photosynthesis, plant pH homeostasis (as ammonia is basic) and the ability to absorb water and nutrients from the soil³. Therefore, similar to how glucose is converted to starch for storage, nitrogen is converted to urea: a stable, neutral compound that can be easily translocated throughout the plant⁴.

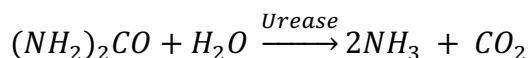
Figure 1: Chemical structure of urea



Urea, either formed as a metabolic waste product or to store nitrogen must be converted back to ammonia so it can be utilized by plants. Ammonia is the building blocks of life, supplying nitrogen in macromolecules like proteins (e.g., chlorophyll to help capture sunlight or enzymes) or nucleic acids (e.g., DNA or RNA).

Therefore, the conversion from urea to ammonia is essential to providing cells with the necessary molecules for them to carry out their function.

Most plants use the enzyme urease to hydrolyze urea into ammonia and carbon dioxide, a reaction that naturally takes 2-4 days but occurs within minutes with urease⁶:



¹ Crop Nutrition. (2023, October 23). Nitrogen | Key nutrients | Mosaic Crop Nutrition. Mosaic Crop Nutrition. <https://www.croptonutrition.com/nutrient-management/nitrogen/#:~:text=NITROGEN%20IN%20PLANTS,-Healthy%20plants%20often&text=Nitrogen%20is%20so%20vital%20because,the%20building%20blocks%20of%20proteins.>

² The Editors of Encyclopaedia Britannica. (2009, February 6). Nitrogen-fixing bacteria | Definition & Types. Encyclopedia Britannica. <https://www.britannica.com/science/nitrogen-fixing-bacteria>

³ Liu, Y., & Von Wirén, N. (2017). Ammonium as a signal for physiological and morphological responses in plants. *Journal of Experimental Botany*, 68(10), 2581–2592. <https://doi.org/10.1093/jxb/erx086>

⁴ Crop Nutrition. (2023b, December 19). *The Facts: Nitrogen fertilizer* | Mosaic Crop Nutrition. Mosaic Crop Nutrition. <https://www.croptonutrition.com/resource-library/the-facts-nitrogen-fertilizer/>

⁵ UREA - American Chemical Society. (n.d.). American Chemical Society. <https://www.acs.org/molecule-of-the-week/archive/u/urea.html>

⁶ Wikipedia contributors. (2024, July 3). Ammonia volatilization from urea. Wikipedia. https://en.wikipedia.org/wiki/Ammonia_volatilization_from_urea

Urease is a catalyst with a globular protein with an active site of the same shape as its substrate (in this case urea) that increases the rate of reaction while remaining unchanged in the process. Urease being an intracellular and extracellular enzyme provides cells with nitrogen, and therefore the maintenance of cellular health is determined by the rate of reaction for urease⁷.

Generally, enzyme reactions are affected by numerous factors, substrate and enzyme concentration, temperature, pH and even the presence of inhibition. However, one of the most regulated and affectable factors is concentration⁸.

Observing the rate of reaction can be challenging, but liquid encapsulation— where a liquid is coated in a semi-permeable membrane— eases this process. This method encapsulates urease in small spheres while still being able to interact with its external environment due to the semi-permeability of the membrane. This method ensures the gradual release of urease into the solution allowing for a uniform interaction with urea. As urease breaks down urea, carbon dioxide gas bubbles are produced which reduce the density of the encapsulated enzyme causing the sphere to rise. This time taken for the sphere to rise directly relates to the rate of reaction.

For experimental purposes, jack beans (*Canavalia ensiformis*) an organism containing high concentrations of urease was used.

Therefore, this leads to my research question: How does the concentration of urease affect the rate of reaction between urease from *Canavalia ensiformis* and urea?

Hypothesis and Null Hypothesis

It was hypothesized that the rate of reaction will initially increase proportionally with an increase in concentration. This is explained by the kinetic particle theory, which suggests that the increasing number of particles per unit area, increase the frequency of collisions between the enzyme and the substrate. This forms more enzyme-substrate complexes and thus produces more product.

The null hypothesis would state that there will no relationship between the concentration of urease and the rate of reaction.

⁷ Jiang, D., Jiang, N., Jiang, H., & Chen, L. (2022). Urease inhibitors increased soil ureC gene abundance and intracellular urease activity when extracellular urease activity was inhibited. *Geoderma*, 430, 116295. <https://doi.org/10.1016/j.geoderma.2022.116295>

⁸ Libretexts. (2023, July 12). 14.3: Effect of Concentration on Reaction Rates: The Rate Law. Chemistry LibreTexts. [https://chem.libretexts.org/Bookshelves/General_Chemistry/Map%3A_General_Chemistry_\(Petrucchi_et_al.\)/14%3A_Chemical_Kinetics/14.03%3A_Effect_of_Concentration_on_Reaction_Rates%3A_The_Rate_Law](https://chem.libretexts.org/Bookshelves/General_Chemistry/Map%3A_General_Chemistry_(Petrucchi_et_al.)/14%3A_Chemical_Kinetics/14.03%3A_Effect_of_Concentration_on_Reaction_Rates%3A_The_Rate_Law)

Methodology

Variables

<i>Independent Variable</i>	<i>Units</i>	<i>Range</i>	<i>Method to Manipulate</i>
Concentration of Urease	g/dm ³	0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5	Exact mass balance was used to measure concentration of urease
		The concentration range was selected to include intervals of 0.05 g/dm ³ which allows accurate identification of the relationship between concentration and rate of reaction. Also, the specific concentrations were based on the concentrations in plant cells to mimic real cellular interactions.	
			Uncertainty of balance $\pm 0.0005 \text{ g/dm}^3$

<i>Dependent Variable</i>	<i>Units</i>
Rate of Reaction	Seconds
	A stopwatch measured the time it took for enough carbon dioxide gas to form, which would have reduced the sphere's density until it floated
	Uncertainty pf stopwatch $\pm 0.005\text{s}$

<i>Control Variables</i>	<i>How it was controlled</i>	<i>Impact on Reaction</i>
Size of sphere	Each sphere was created with 2 mL of urease solution	Variations in the size and shape of the spheres could potentially increase the rate of reaction for larger spheres or decreasing it for smaller ones exposing more/less surface area to the urea. Moreover, the apparatus used was constant throughout the experiment, therefore despite uncertainties associated with the equipment, all the results would have the same level of inaccuracy.
Shape of sphere	Abnormally shaped spheres were removed before the experiment	
Temperature of urea	All the experiments were conducted at room temperature (37°C)	If the temperature wasn't controlled for, the rate of reaction would increase and decrease with higher and lower temperatures respectively.
Age of <i>Canavalia ensiformis</i>	The experiment was conducted within an hour of preparing the fresh solution	As the jack beans age, the activity of the urease may result in a decrease in enzyme activity ⁹ . Therefore, considering that the experiment measures the effect of temperature on rate of reaction, it was crucial to ensure the age of the urease didn't affect the results.
Concentration of urea	A 0.1M of urea solution was used	If the concentration/volume of hydrogen peroxide were to be higher, this would result in faster reactions, and vice versa for lower concentrations despite the temperature changes. Furthermore, despite the uncertainties associated with the equipment, the same apparatus was used throughout the experiment thus ensuring all the results contain the same level of inaccuracy.
Volume of urea	20mL of hydrogen peroxide was used for each trial	

⁹ Pawar, V. S., Bhande, D., Pawar, S. D., Mudila, H., Kaushik, A., & Kumar, A. (2022). Investigating purification and activity analysis of urease enzyme extracted from jack bean source: A green chemistry approach. *Analytical Biochemistry*, 659, 114925. <https://doi.org/10.1016/j.ab.2022.114925>

<i>Confounding variable</i>	<i>Impact on Reaction</i>
pH of urease/pH of urea	The pH wasn't measured during the experiment. Any deviation from ureases optimal pH of 7.5 could lower the rate of reaction ¹⁰ .
Room temperature fluctuations	Slight variations in the temperature could affect the kinetic energy, thus increasing/decreasing the rate of reaction accordingly.
Humidity and moisture fluctuations	Higher humidity would result in dilution of the urease solution thus decreasing the rate of reaction and vice versa for a reduction in humidity ¹¹ .
Dissolved carbon dioxide levels	Higher levels of dissolved carbon dioxide in the urease solution could trigger the reaction before the experiment began.
Thickness of encapsulated layer	Differences in the alginate coating could affect the diffusion of urease in and out of the sphere.

Apparatus

<i>Equipment</i>	<i>Uncertainty</i>
0.1M urea solution	
Urease powder	
Sodium alginate powder	
0.15M calcium chloride solution	
Measuring cylinder	±0.5mL
Thermometer (0-110°C)	±0.5°C
100mL Beaker	
30mL Syringe	
Stirring rod	
Tweezers	
Strainer	
Distilled water	
Graduated measuring cylinder	±0.05mL
Stopwatch	±0.005s
Balance (0-100 g)	±0.005g

¹⁰ Wikipedia contributors. (2025, May 23). *Urease*. Wikipedia. <https://en.wikipedia.org/wiki/Urease>

¹¹ Lindeque, R. M., & Woodley, J. M. (2020). The Effect of Dissolved Oxygen on Kinetics during Continuous Biocatalytic Oxidations. *Organic Process Research & Development*, 24(10), 2055–2063. <https://doi.org/10.1021/acs.oprd.0c00140>

Safety, ethics and environmental issues

<i>Substance</i>	<i>Danger</i>	<i>Solution</i>
Urea	Safety issue – urea can cause irritation/burns if it on skin/eye and respiratory irritation ¹² when inhaled. Environmental – improper disposal could contribute to nitrogen pollution	Wear gloves and eye-protection and avoid inhalation. Waste was disposed in accordance with school's lab safety protocol
Urease	Ethical issue – improper disposal could contaminate biological ecosystems Safety – inhalation and contact with eyes can cause irritation	Dilute the urease with water and lots of water and dispose down the drain. Wear gloves and eye protection when handling to avoid infection.

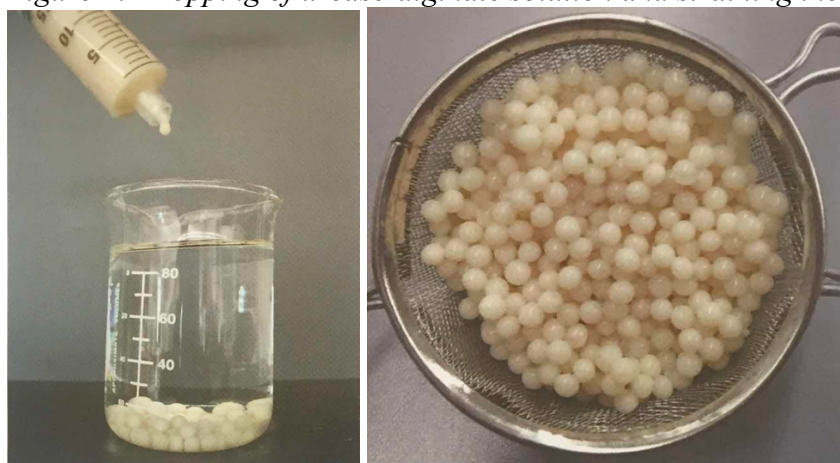
Method

Method to prepare urea-alginate spheres

1. Created 0.1M urea solution
 - a. Measured 1000mL of distilled water with a measuring cylinder
 - b. Added to 6g of jack bean powder measured with a balance
 - c. Mixed these and stir
2. Prepared urease solutions (0.05, 0.1, 0.15, 0.2, 0.25, 0.3 g/dm³)
 - a. Used a graduated measuring cylinder to measure 30mL of water
 - b. Added it to the according concentration of urease
 - c. Stirred with a stirring rod
3. Prepared 2% sodium alginate solution
 - a. Measured 100mL of distilled water with a measuring cylinder
 - b. Added this to 2 grams of sodium alginate power weighted on a balance
 - c. Stirred with a stirring rod
4. Created 0.15 M CaCl₂
 - a. Measured 100mL of distilled water with a measuring cylinder
 - b. Add it to 1.67 grams of CaCl₂ measured with a balance
5. Combined 100mL of the urease solutions with 100mL of 2% sodium alginate solution.
6. Using a 30mL syringe (± 0.5 mL) to draw up sodium alginate, slowly released droplets into the 0.15 CaCl₂ M solution forming spheres
 - a. See figure 2

¹² CF Industries. (2024). Urea, granular Safety Data Sheet. <https://www.cfindustries.com/globalassets/cf-industries/media/documents/safety-data-sheets/granular-urea---north-america/granular-urea-english.pdf>

Figure 2: Dropping of urease-alginate solution and straining them¹³



7. Leave spheres for 5 minutes in CaCl_2 to harden before straining them
8. After the syringe is empty and repeat step 6-8 to refill the syringe and create more spheres
9. Dispose of any floating spheres in the solution of CaCl_2
10. When all spheres are created strain CaCl_2 solution and rinse the spheres with water to prevent drying
11. Repeat using the different concentrations of urease



Figure 3: Actual spheres made

Method: Testing effect of concentration on rate of reaction

1. Using an accurate measuring cylinder (uncertainty 0.05mL) transfer 30mL of urea solution to a test-tube
2. Start the stopwatch ($\pm 0.005\text{s}$) when the sphere is added to the test-tube by tweezers
3. Stop the stopwatch ($\pm 0.005\text{s}$) when the sphere has fully floated to the top of the test-tube
4. Repeat steps 1-3 for concentration of urease
5. Repeat steps 1-4 five times such that five trials are calculated for each concentration
 - a. This ensures the results are as reliable as possible and reduces the likelihood of errors

¹³ The Science Teacher September 2016: A twist on measuring catalase. (n.d).
http://onlinedigitalpublishing.com/publication/?i=327538&article_id=2554129&view=articleBrowser

Data Presentation

Raw Data

Data Table 1: A table showing how concentration of urease affects the time for the sphere to rise

Concentration\ g/L (± 0.5 g/L)	Time for Sphere to Rise / s (± 0.05 s)					Average	Standard Deviation
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5		
0.05	72.88	45.12	66.01	53.34	57.68	59.01	10.82
0.10	33.75	42.79	48.92	35.21	39.13	39.96	6.13
0.15	27.10	22.31	24.90	21.59	28.77	24.93	3.06
0.20	16.90	13.25	9.77	11.31	13.10	12.87	2.67
0.25	13.38	13.70	11.58	14.96	12.07	13.13	0.76
0.30	10.82	11.64	12.18	11.31	11.58	11.51	0.50
0.35	10.45	11.21	10.32	11.06	10.97	10.80	0.39
0.40	10.01	10.39	10.78	10.76	10.13	10.41	0.35
0.45	10.32	9.81	10.58	9.86	10.27	10.17	0.33
0.50	9.84	10.13	10.01	10.29	10.12	10.08	0.17

Qualitative Observations

When the spheres were placed in the calcium chloride, some bubbles/fixing was visibly seen. Although the fizz was relatively random, generally, as the concentration of urease rose, the fizzing increased until the concentration of urease reached 0.35 g/L where the fizzing remained constant. Not only was effervescence qualitatively measured, but this reaction released some heat, and at higher concentrations there was more heat generated.

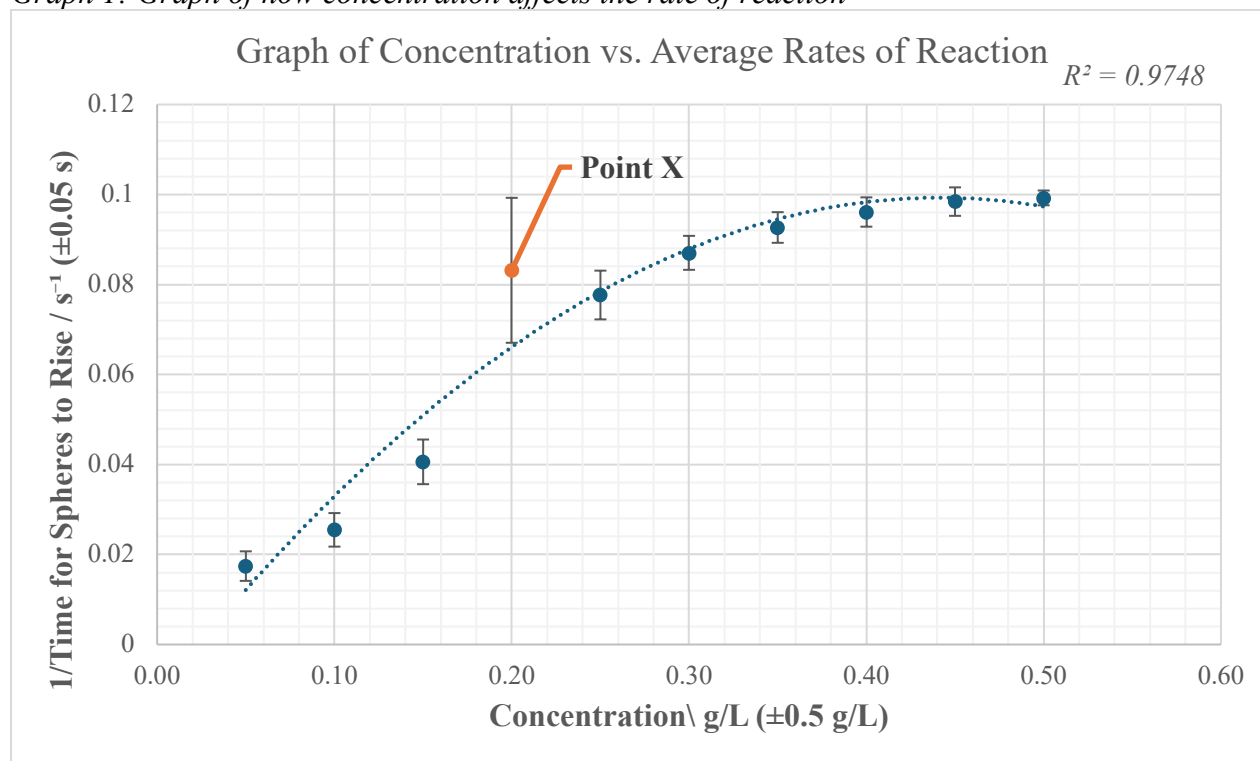
Processed data

To calculate the rate of reaction, the equation $\frac{1}{t}$ was used, where t was the time for sphere to rise. This was used to convert the raw time-data into rates of reaction. The standard deviation was calculated using the equation: $\sqrt{\frac{\sum(x_i - \mu)^2}{N}}$ where Σ = sum of, x_i = each value in the data set, μ = mean of all values in the dataset, and N = number of values in the dataset.

Data Table 2: A table showing how concentration affects the rate of reaction

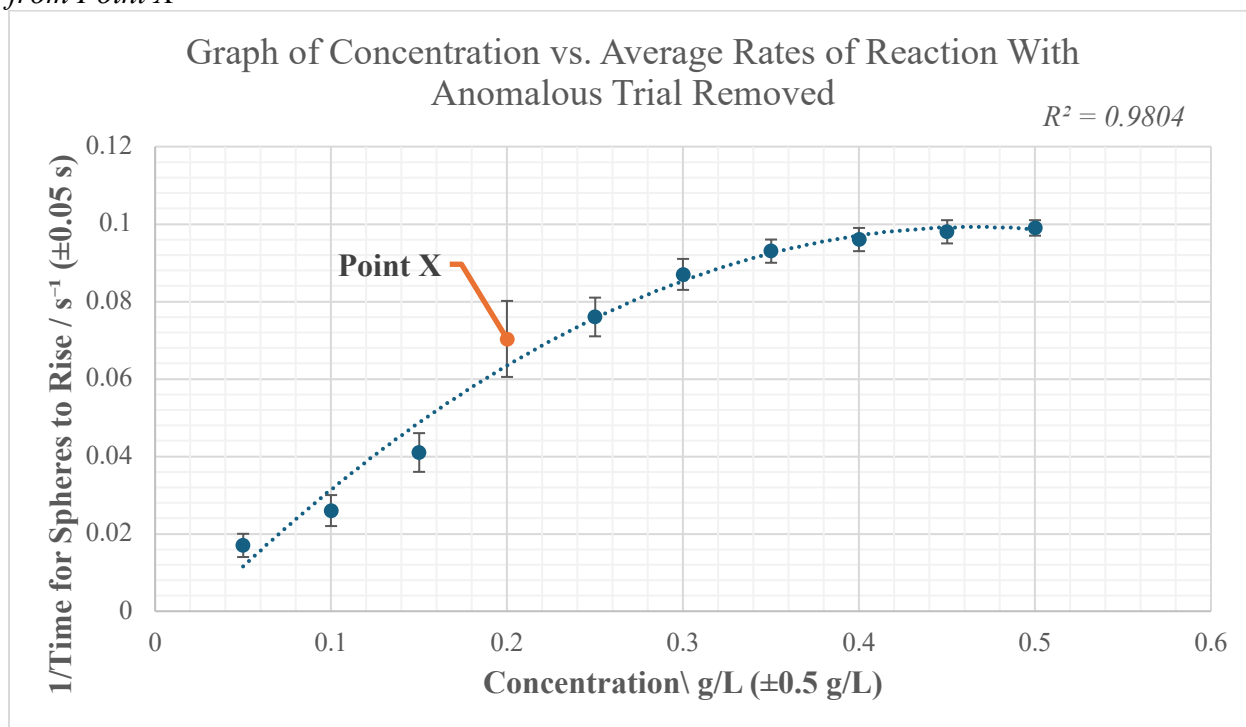
Concentration\ g/L (± 0.5 g/L)	1/Time for Spheres to Rise / s^{-1} (± 0.05 s)					Average	Standard Deviation
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5		
0.050	0.014	0.022	0.015	0.019	0.017	0.017	0.003
0.100	0.030	0.023	0.020	0.028	0.026	0.026	0.004
0.150	0.037	0.045	0.040	0.046	0.035	0.041	0.005
0.200	0.059	0.076	0.102	0.088	0.076	0.080	0.016
0.250	0.080	0.078	0.085	0.091	0.080	0.076	0.005
0.300	0.092	0.086	0.082	0.088	0.086	0.087	0.004
0.350	0.096	0.089	0.097	0.090	0.091	0.093	0.003
0.400	0.100	0.096	0.093	0.093	0.099	0.096	0.003
0.450	0.097	0.102	0.095	0.101	0.097	0.098	0.003
0.500	0.102	0.099	0.100	0.097	0.099	0.099	0.002

Graph 1: Graph of how concentration affects the rate of reaction



Point X has a large standard deviation being ± 0.0161 which emphasizes the variability for the data collected at 0.2 g/L. The rate of reaction also higher than expected from the trendline. Thus, Point X doesn't follow this trend and is thus an anomaly. Specifically, Trial 3 and 4 were higher than expected, and thus in the following graph these trials were removed when calculating the Average and Standard Deviation of Point X.

Graph 2: Graph of How Concentration Affects Rate of Reaction After Removing Trial's 2 and 3 from Point X



Removing those anomalous Trials seemed to reduce the error and ensure the point fits with the overall trend.

Table 3: Table outlining calculation of sum of least square residuals

Concentration\ g/L (±0.5 g/L)	Actual Rate of Reaction	Predicted Rate by Trendline	Residual	Residual Squared
0.050	0.017	0.005	-0.013	0.000
0.100	0.025	0.031	0.006	0.000
0.150	0.041	0.060	0.019	0.000
0.200	0.083	0.091	0.008	0.000
0.250	0.078	0.125	0.047	0.002
0.300	0.087	0.161	0.074	0.005
0.350	0.093	0.199	0.107	0.011
0.400	0.096	0.240	0.144	0.021
0.450	0.098	0.284	0.185	0.034
0.500	0.099	0.329	0.230	0.053
Sum of least square residuals is 0.128				

The sum of squared residuals was calculated using the formula $\sum_{i=1}^n (y_i - f(x_i))^2$, where Σ = sum of, n is the upper limit, y_i is the actual rate of reaction and $f(x_i)$ is the predicted value by the trendline. The closer this value to 0, the better the fit of the trendline.

The R^2 coefficient was calculated by the equation $1 - \frac{RSS}{TSS}$ where RSS was the sum of squares of residuals and TSS was total sum of squares. The R^2 coefficient was 0.984

A one-way ANOVA (Analysis of Variance) was computed to compare the means of each rate of reaction for each concentration and determine if there is a statically significant difference between them. This test was suitable as the data had a normal distribution and because the independent variable was categorical (different concentrations) and the dependent variable continues (rate of reaction) the ANOVA test was used.

ANOVA was calculated using the equation $\frac{SS_{Between}}{\frac{k-1}{N-k}}$, where $SS_{Between}$ was the variation between group means, SS_{Within} , was the variation within groups, k was the number of groups and N was the total number of observations. When conducted, a p-value of >0.001 was identified, highlighting that the enzyme concentration had a statistically significant effect on the rate of reaction.

Conclusion

In conclusion, as the concentration increased from 0.05 g/L to 0.3 g/L the rate of reaction of urease increased, and from 0.3g/L to 0.5 g/L, the rate of reaction began to plateau. The data collected in the investigation mostly supports the previously stated hypothesis where rate of reaction increases as concentration increases due to kinetic particle theory. At low enzyme concentration, the enzyme itself is the limiting factor as there are many more substrate molecules than enzyme, and thus an increase in the concentration of an enzyme exponentially increases the rate of reaction. However, at higher enzyme concentrations, the substrate becomes the limiting factor due to the limited availability of substrate molecules, and an increase in the enzyme concentration doesn't drastically increase the rate of reaction and it begins to plateau.

From the raw data collected in Table 1 the average duration of the reaction decreased exponentially changing from an average of 59.02 seconds when 0.05g/L of urease was used to 12.87 seconds when 0.2 g/L of urease was used. This time began to plateau, changing only by 2 seconds from 0.25 g/L to 0.5 g/L of urease. This pattern is further justified through the qualitative observations, as there was minimal effervescence when low concentrations of urease were used (0.05-0.1g/L), but this effervescence increased, until it remained constant after 0.35 g/L. From the processed data in Table 2, the rate of reaction was calculated, where initially, from 0.05g/L to 0.35g/L there was a positive correlation between concentration and the rate of reaction, and after, this, the rate of reaction remained relatively constant.

Graph 1 visually displays the trend outlined in Table 2 and the error bars represent standard deviation, which remained mostly negligible ranging from 0.002 to 0.0054 excluding point X. Point X had the highest standard deviation being 0.0161 which was significant. Furthermore, based on scientific consensus, at 0.2 g/L, the rate of reaction should have been relatively low. The low concentration meant that few active sites of the enzymes were readily available for substrates to bond with and thus would have reduced the enzyme-substrate complexes formed.

However, the rate was 0.0802 which was higher than expected. Possibly the immense error at point X explains the lack of conformity to the trend highlighting point X as an anomaly. Finally, as the reaction increased the temperature, the gas released may have expanded, which artificially created higher rates of reaction which also explains the anomaly X although interestingly it wasn't observed in the other trials.

Generally, Graph 1 highlights the R^2 coefficient to be 0.984 which is close to 1, signifying that the polynomial trendline chosen accurately models the relationship. Table 3 summarizes the least square residual, which further describes the fit of the trendline to the data present. A value of 0.128 highlights that the squared sum of all the differences between the predicted Y-value from the trendline and the actual points were minimal further justifying the specific trendline for the data. Finally, the low p-value of <0.001 was determined, which highlights how there is a less than 0.1% chance that the differences in the reaction rate occurred by chance. This means that the null hypothesis can be rejected. Both statistical measures support the hypothesis as they justify the pattern of the effects of concentration on enzyme activity.

In the scientific community, Sehgal and Naylor (1966) purified jack bean urease and observed that there was a typical saturation of rate of reaction as the concentration increases, which aligns with the findings of this study. Sehgal and Naylor (1966) also highlighted how even among the same species of *Canavalia ensiformis* there is high variation in the concentration of urease. However, Dr. Fay's justification for the trend of enzyme concentration and rate of reaction is like the hypothesis, stating how an increase in concentration, increases the availability of the three-dimensional active sites of enzymes, producing more product. In first-order reactions (which this reaction is) the rate of reaction is initially directly proportional, which is due to the specific Arrhenius factor. Also, the paper highlights how the system approaches V_{max} as the substrate becomes the limiting factor which illustrates the plateau in reaction rates.

Evaluation

Positive	Source of error	Solution
The trend/pattern in the results are clear as the standard deviation and sum of least squares were low, with a very strong positive R^2 . This allowed the RQ to be accurately assessed.	There were some anomalies, and the specific mixture of jack beans wasn't homogenous, as after a few minutes the urease would separate from the water which possibly caused anomaly X. This prevented the RQ from being thoroughly validated as the differences in reaction could have arisen from the composition of urease.	Continue to mix the solution when before taking samples from the syringe
The range of independent variable values (concentration) produced enough data to sufficiently answer the research question as 10 different concentrations were tested allowing the accurate assessment of the RQ.	Urease would naturally decompose. Thus, this affected the ability for the RQ to be accurately assessed.	Conduct a negative control, where no urea is added to see how the natural decomposition of urease affects the density of the spheres
The equipment used where the measuring cylinder ($\pm 0.05\text{mL}$) and stopwatch ($\pm 0.005\text{s}$) were precise enough to prevent significant errors in the results thus validating the RQ.	The pH of urease or urea wasn't factored into the experiment. Urea is slightly basic and thus could have artificially reduced the rate of reaction and thus prevented the RQ from being accurately assessed. ¹⁴	Use a buffer solution to ensure that the pH of the solution is constantly at 7 which is the optimum for <i>Canavalia ensiformis</i>
The dependent variable (the rate of reaction) was measured and produced valid results	Some random errors affect this measurement such as the reaction time of the individual to stop the stopwatch or the randomness of enzyme-substrate reactions created variations between the different trials. However, this didn't significantly affect the ability to answer the RQ.	More trials could have been conducted to reduce the effects of random errors. This would have allowed the RQ to have been better evidenced.
	A key systematic error were the environmental conditions weren't kept constant; the environmental temperature and humidity would drastically fluctuate which would have had a large impact on the results. This significantly affected the ability for the RQ to be answered.	This experiment could have been done at standard conditions of 20°C with humidity set to 50% by a humidifier to reduce the effect of systematic errors. This would have resulted in a more accurate assessment of the RQ.

¹⁴ Wikipedia contributors. (2025b, May 25). Urea. Wikipedia. <https://en.wikipedia.org/wiki/Urea>

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